Leopold Wuhan Zhou

Beijing, China | wuhanzhou@stu.pku.edu.cn | leopoldwuhanzhou.github.io

REASEARCH INTEREST

- **Combinatorics, especially algebraic combinatorics**
- **Mathematical physics, especially those problems related to combinatorics**
- **Applied mathematics, related to AI and financial math**

EDUCATION

2023 - Present **B.S. in Electronic Information Science and Technology**, Peking University
2023 - Present **B.S. in Mathematics**, Peking University
Rank: 8

SELECTED COURSES

Mathematics: Algebra, ODE, Complex Analysis, Differential Manifolds, Topology, Real Analysis, Groups and Representations, Lie Groups and Lie Algebras, Probability, Stochastic Processes
Physics: Quantum Mechanics, Electrodynamics, Statistical Physics
EECS: Introduction to AI, Signals and Systems, Analog Circuits, Digital Circuits, Optoelectronics

PAPERS

1. *Realizations and Uniqueness of Cut Complexes of Graphs.* 2025 (With Yufeng Shen, Zhiyu Song, Fenglin Yu, Jingqi Zhuang) arXiv:2512.12933 Accepted by Annals of Combinatorics
2. *Neighborhood Complexes of Induced k-independent Graphs.* 2025 (With Yufeng Shen, Zhiyu Song, Fenglin Yu, Jingqi Zhuang) arXiv:2512.09674
3. *Homotopy Type of Total Cut Complexes of Squared Cycle Graphs.* 2025 (With Yufeng Shen, Zhiyu Song, Fenglin Yu, Jingqi Zhuang) arXiv:2512.09674

EXPERIENCES

Summer 2026 **Mathematics REU**, University of Chicago

- My mentor is Aman Panigrahi.
- I studied the performance of classical greedy algorithm and RL on maximal independent set probelm.

January 2026 - October 2026 **REU on Stokes Phenomenon**, Peking University

- My mentor is Xiaomeng Xu.
- I, with Yuxuan Cai, studied Stokes phenomena in meromorphic differential equations and the construction of quantum group representations from Stokes matrices via RLL relations.

Summer 2025 **PKU Algebraic Combinatorics Experience (PACE)**, BICMR

- I collaborated with Yufeng Shen, Zhiyu Song, Fenglin Yu and Jingqi Zhuang, and our mentor is Lei Xue. This REU is organised by Yibo Gao.
- We studied cut complexes and total cut complexes. Especially, we figured out the homotopy type of squared cycles in some cases. We investigated on the connection between total cut complexes and induced neighborhood complexes. Also, we found out the realizations and uniqueness of cut complexes of some graphs.

Summer 2024 **AI for Mathematics Summer Camp**, BICMR

- This summer camp is organised by Liang Xiao and Bin Dong.
- We learned abstract algebra and wrote Lean4 programs to help AI models to solve math problems.

CONFERENCES & SEMINARS

August 10th 2026 International Congress of Basic Sciences (**ICBS**) Poster Session

- Total Cut Complex and Their Homotopy Types.

Spring 2026 **Algebra and Number Theory** organized Liang Xiao

- Hilbert Symbol.

Fall 2025 **The Integrability Problems in Mathematical Physics** organized by Xiaomeng Xu

- Yang-Baxter Equation and its quantum mechanical origin.
- Yang-Baxter Equation and braid groups.

AWARDS & SCHOLARSHIPS

2026 China Merchants Securities Scholarship, Peking University
Award for Scientific Research, Peking University
2025 Zhang Naixin Scholarship, Peking University
Award for Scientific Research, Peking University
2024 Yang Xin Lotus Virtue Awards, Peking University
Award for Academic Excellence, Peking University

SERVICE & LEADERSHIP

Fall 2024-Spring 2025 **Vice Minister of Academic Department of Student Union**, School of EECS
Fall 2023–Spring 2024 **Member of Academic Department of Student Union**, School of EECS
Member of the Youth Volunteer Association, School of EECS

SKILLS

Programming language C/C++, Lean4, Latex, Matlab, Python, SageMath, Verilog
Language Chinese, English (TOEFL 5.5/6.0), Shanghainese, Italian (Learning)